Run report: ctrlH-q5-run-20261024T2156Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part HYPER under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: ctrlH-q5 20261024T2156Z
git HEAD: 0fb870d0ee76ba6a1b059e9e34c869518fb412b0
tool: examples/genus4_n4_census.rs sha256 f4c7f7355c838b6f112e09cb412fd3353d087309962c64cd39ad49d79f4c9dad
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 a95b1e35748cefe23a044d8cfd9e059b723cf71fdbc4af56db71257da42c8b33
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/ctrlH-q5-run-20261024T2156Z HYPER 5 50000000 202610245
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 95
```

tool_log.txt

```
HYPER q=5 mode=sampled forms_examined=50000000
squarefree_degree10_forms=38397517
res_shaped=119496
odd_t_hits=0
t_hist={-50: 184, -46: 477, -44: 528, -36: 989, -34: 7507, -32: 2461, -30: 960, -28: 1037, -26: 1444, -24: 4309, -22: 8287, -20: 1435, -18: 3420, -16: 1916, -14: 17066, -12: 2030, -10: 5422, -6:
1971, -4: 10316, -2: 537, 0: 1157, 2: 8590, 4: 1497, 6: 4420, 10: 3041, 12: 1967, 14: 7564, 16: 461, 18: 3713, 20: 1921, 22: 2008, 24: 529, 26: 5424, 32: 2148, 34: 497, 36: 489, 38: 971, 46:
604, 50: 199}
seconds=94.2
```

summary.txt

```
HYPER q=5 mode=sampled forms_examined=50000000
squarefree_degree10_forms=38397517
res_shaped=119496
odd_t_hits=0
t_hist={-50: 184, -46: 477, -44: 528, -36: 989, -34: 7507, -32: 2461, -30: 960, -28: 1037, -26: 1444, -24: 4309, -22: 8287, -20: 1435, -18: 3420, -16: 1916, -14: 17066, -12: 2030, -10: 5422, -6:
1971, -4: 10316, -2: 537, 0: 1157, 2: 8590, 4: 1497, 6: 4420, 10: 3041, 12: 1967, 14: 7564, 16: 461, 18: 3713, 20: 1921, 22: 2008, 24: 529, 26: 5424, 32: 2148, 34: 497, 36: 489, 38: 971, 46:
604, 50: 199}
seconds=94.2
```

hyper_t_hist.csv (first 12 of 39 rows)

t	count
−50	184
−46	477
−44	528
−36	989
−34	7507
−32	2461
−30	960
−28	1037
−26	1444
−24	4309
−22	8287
−20	1435

Graph

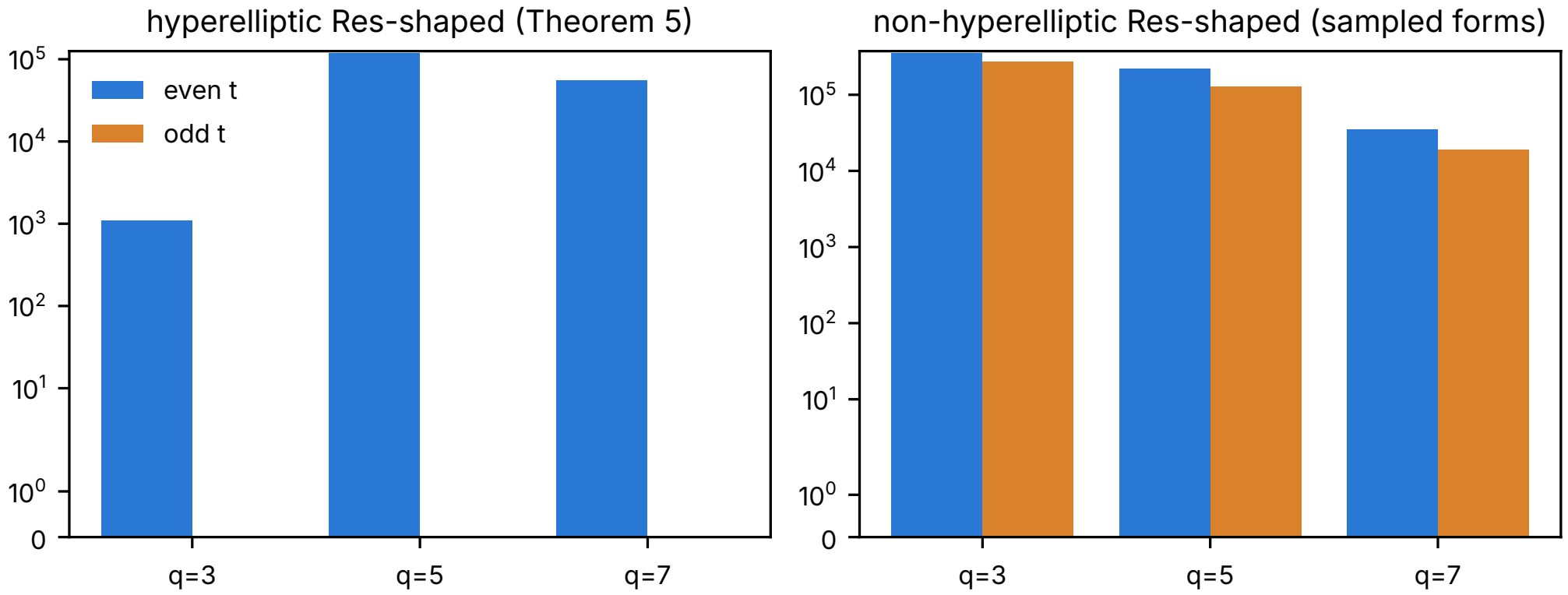

Figure 1: Parity of Res-shaped traces: hyperelliptic (Theorem 5, no odd t) and non-hyperelliptic.